Smart Scan

Lisa Deschler
l.deschler@campus.lmu.de
Ludwig-Maximilians-Universität München
München, Bayern, Deutschland

Amy Grindau
Ludwig-Maximilians-Universität München
München, Bayern, Deutschland

Gaman Thach
g.thach@campus.lmu.de
Ludwig-Maximilians-Universität München
München, Bayern, Deutschland

Louis Zimmermann
Ludwig-Maximilians-Universität München
München, Bayern, Deutschland

Abstract

CCS Concepts

• **Hardware → Scanners**.

Keywords

Assistive Technology, Artificial Intelligence, Older Adults, Prototyping, High-Fidelity, Low-Fidelity

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1 Introduction (Louis Zimmermann)

The "Design Workshop 2" seminar, held in SoSe 2025, focused on the overarching theme of "Assistive AI - Lowering barriers through intelligent UX Design Solutions." This theme underscores the increasing importance of designing technology that genuinely supports users, particularly vulnerable groups, without imposing unnecessary complexities or disrupting established routines. As Bill Buxton aptly stated, "UX Designers not only tell a story, they listen to one," highlighting the crucial role of user-centered design in developing meaningful solutions. Our project aimed to explore how Artificial Intelligence (AI) can serve as an assistive tool for the elderly, addressing their specific needs and concerns while navigating their often critical stance towards new technologies that might interfere with their daily lives. This report details the initial phases of our design process, encompassing the comprehensive research and ideation stages that led to our proposed solution, SmartScan.

2 Research Phase

Our research phase commenced with an in-depth exploration of the current landscape of "Assistive AI for the Elderly." This involved a thorough desk research, examining existing studies and products available on the market. The goal was to understand prevalent challenges faced by the elderly concerning technology adoption and the potential of AI to mitigate these issues.

Following the desk research, we proceeded with qualitative user research through interviews with real individuals from our target demographic, specifically those aged 74 and above. Participants included individuals with varying degrees of technological experience, from those who exclusively used landline phones to others who were proficient smartphone users. The interviews focused on their general technology usage habits, their perceptions of AI, and their willingness to integrate new digital tools into their routines. A key finding from these interviews was a significant skepticism towards AI, often expressed as a preference for self-reliance and a distrust of automated systems, as one participant articulated: "[...] I must honestly confess that I don't trust an AI."

The insights gathered from these interviews were meticulously analyzed to develop detailed personas, outlining the diverse technological backgrounds and attitudes within our target group. We identified key "Pains" (frustrations and challenges) and "Gains" (desired benefits and positive experiences) related to technology use among the elderly. This analysis culminated in four main insights:

- Technology is accepted when it simplifies, not when it interferes with the self-evident.
- Lack of knowledge about AI creates uncertainty and rejection, but not necessarily technophobia.
- User-friendliness and understandable design are central requirements for technical products, especially regarding usability.
- Technology must not replace existing routines but must meaningfully supplement them.

These insights underscored a critical problem: elderly individuals often react critically to the introduction of anything unnecessary into their daily routines. Any proposed solution needed to be non-intrusive and demonstrably beneficial.

3 Ideation Phase

With a clear understanding of our target group's needs and reservations, we transitioned into the ideation phase. This stage involved employing various brainstorming and idea selection methods to generate potential solutions that aligned with our research findings. We utilized techniques such as the 6-3-5 Brainwriting Method, which encouraged participants to build upon each other's ideas in a structured manner, and the COCD Box, a tool for categorizing ideas based on their feasibility and originality. Throughout this process, our primary focus remained on addressing the identified

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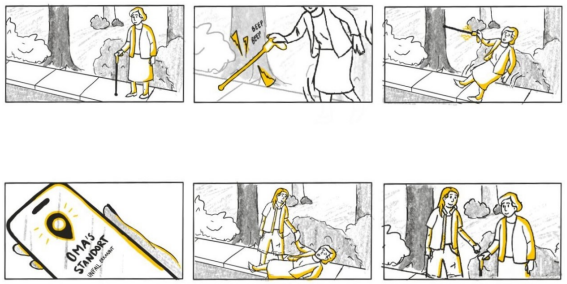

Figure 1: Storyboard Smart Walking Cane

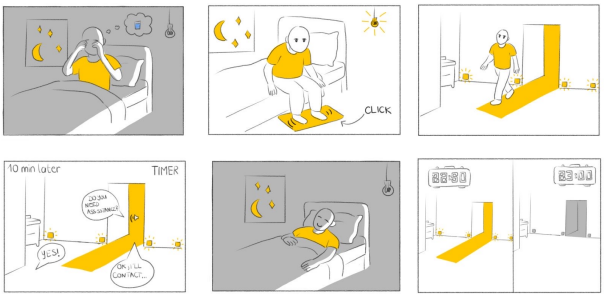

Figure 2: Storyboard Floor Mat

Figure 3: Storyboard Smart Scan

problem: developing a solution that genuinely solves a problem without adding unnecessary complexity or disrupting established routines for the elderly. This rigorous ideation process ultimately led to the development of our core concept: SmartScan.

4 Design process (Gaman Thach)

4.1 Storyboards

Before we finalized our idea, we chose the three most promising results that emerged during the ideation phase: a smart walking cane, a light mattress and a smart scanner. For each product idea, we brainstormed what core functionalities are needed to fulfill the needs of potential end users. In addition, each product will be supported by an AI system. Based on these core functionalities, we created three different storyboards that showcase each product idea.

4.1.1 Smart walking cane (Figure 1). An ordinary walking cane with additional technological features such as a warning system for obstacles that use vibrations to alert the user and an emergency feature with location tracking in case of emergencies. We wanted to highlight an ergonomic design for comfort and stability. The main goal of this product idea is to ensure security and independence of older people as much as possible.

4.1.2 Floor Mat with Light-System (Figure 2). A mat that is connected to an intelligent light system in a home environment. It recognizes movements and daytime to optimize home light usage and convenience by automatically turning lights on and off, depending on daytime. Enables light switching without the need to manually turn a light switch off or on. It saves energy consumption because of higher optimization rates and increases security, comfort and independence in the daily lives of older people.

4.1.3 Smart Scan (Figure 3). A device that scans user instructions and other potentially hard to read or difficult to understand wordings to reinterpret it in easier to understand terms. It utilizes visual and speech-based approaches to simplify texts for end users. Additionally, it emphasizes accessibility through text-to-speech functionality. Its main goal is to let anyone profit from technological inventions.

After careful consideration and discussion, we concluded that the smart scanner was the most feasible solution, as it fit within our time and technical resource limitations.

4.2 Refinement

Once we settled on one idea, we started brainstorming about the design of the product. For that, we made first simple sketches and used an AI image generator to give us first design approaches. We concluded that the product should have a simple familiar appearance and opted to create a design similar to regular scanners. Then we further refined the key features. SmartScan needs to be able to scan documents, simplify text, contain text-to-speech functionality, include the possibility to display a bigger text size and it has to give feedback after a scan is complete to signal the user, that the other functions of the device are ready as well as a tactile interface

Figure 4: Storyboard of Scan Function

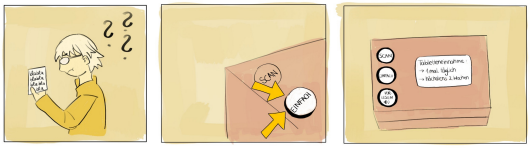

Figure 5: Storyboard of Simplify Function

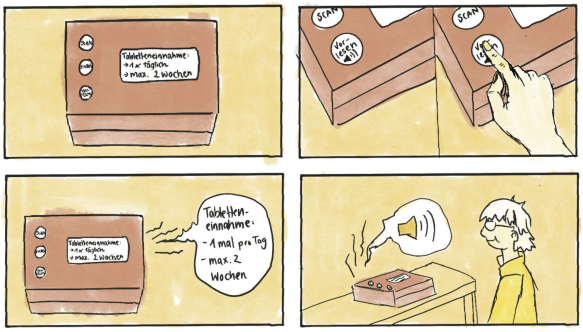

Figure 6: Storyboard of Text-To-Speech Function

with physical components. Afterwards, we compiled the necessary components to realize these features and to build the prototype. Namely, a big screen to depict simplified text in bigger sizes and a speaker to provide text-to-speech functionality, in order to include people with visual or hearing impairments. Furthermore, to scan documents we need to install a camera. Once the documents are scanned, they are sent to an AI model to simplify the text. To ensure proper feedback after a scan is completed, we thought of implementing LED lights to turn on during the scanning process. After the scanning process is completed, the LED lights are turned off. Additionally, each core functionality is activated by physical buttons.

To showcase each key feature in more detail, we created three additionally storyboards, each focusing on one feature. In the first storyboard (Figure 4), the scan functionality is explained with simple instructions. We then see an older person clicking on the scan button on the SmartScan device. The second storyboard (Figure 5) shows a confused elderly person with a document that uses the simply functionality of SmartScan to get an easier to understand text. The final storyboard (Figure 6) shows an elderly person that uses the button that enables the text-to-speech functionality. The device then reads out simplified text to them.

Following consultation with our supervisors, we decided to implement our prototype without the camera, and therefore without the scan functionality. Instead, we hard-coded a sample document, as implementing a camera that could capture documents at close range proved to be difficult.

5 Prototyping and Evaluation (Lisa Deschler)

5.1 Low-Fidelity

For our low-fidelity prototype, we initially focused on the design. Our goal was to ensure that the object could be seamlessly integrated into the user's home without being perceived as intrusive. The design was intended to take the shape of a compact box to save space and correspond to the typical dimensions of paper documents.

We developed two different versions of the prototype:

- Prototype 1 (Open Box): Technology visibly integrated, functionally designed (Figure 1)
- Prototype 2 (Concealed Chest): Lockable, inspired by everyday household items (Figure 2)

Both prototypes were primarily constructed from cardboard. In parallel, we created several button variants for the first prototype to test aspects such as contrast, size, color, and tactile design.

In Prototype 1 (Figure 7), the display was positioned on top of the box. In Prototype 2 (Figure 8), we explored a folding mechanism, where the camera would be mounted on the inside of the lid. In this case, the camera angle would theoretically be adjustable when opening the device. The buttons were also placed inside the box, as well as the display.

Figure 7: Low-Fidelity Prototype 1

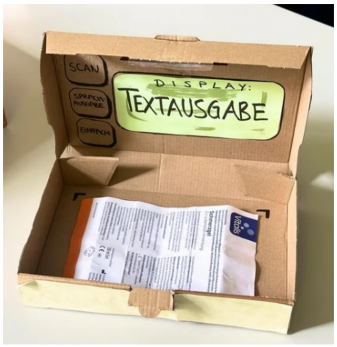

Figure 8: Low-Fidelity Prototype 2

5.1.1 Design Iteration. We conducted individual usability sessions with two people over the age of 70, allowing them to explore both prototypes before conducting follow-up interviews. The key findings were as follows:

(1) Visual Design: Stronger contrasts (e.g., white on black or vice versa) and larger text were needed. Icons were often unclear or misunderstood — for example, the speaker symbol. The overall size of the box was acceptable, though it should not be any larger. Prototype 2 was appreciated for its protective element, as the internal mechanics were theoretically enclosed. However, Prototype 1 was perceived as more straightforward and practical to use. The label for Button 2 ("Sprachausgabe"/"Speech Output") was not entirely clear, and participants emphasized the importance of functionality over aesthetics.
(2) Functionality: The slot for inserting documents was not immediately recognized as such. Participants expressed a desire for a power switch, e.g., to avoid the impression of "spying." In Prototype 2, the adjustable camera angle was perceived as unclear.

Conclusion: Prototype 1 received more positive feedback from the target group. Based on the results of the design iteration, we selected this version and refined it accordingly (Figure 9).

5.2 High-Fidelity

5.2.1 Materials: The high-fidelity prototype featured a housing constructed from laser-cut wood. It included three buttons (Scan, Easy, and Play) that were 3D-printed, along with an LED strip and a display. The box had two wood panels that were mounted with hinges. (Figure 10.

5.2.2 Feedback: The LED strip provides visual feedback when a document or instruction leaflet is inserted. Initially, we attempted to implement this using an ultrasonic distance sensor, but due to technical issues, the LEDs are now activated by pressing the Scan button. The 3D-printed buttons were tactile and provided physical feedback. Labels were applied using high-contrast black-and-white stickers.

5.2.3 Functionality: A web server was implemented using the Raspberry Pi Pico, which allowed the display to be connected to the Raspberry Pi. Once connected, usage instructions appeared on the screen. Since the device did not yet have a functional scanning feature or AI integration, the displayed content was hard-coded.

The buttons were used to access three core functions:

(1) Scan Button: Initiates the scanning process and provides confirmation that the document has been successfully scanned.
(2) Easy Button: Enlarges the scanned document on the first press; on the second press, it displays a simplified/summary version.
(3) Play Button: Plays audio files related to each screen on the display via a speaker.

6 Outlook (Amy Grindau)

SmartScan's significant potential as an assistive AI solution for the elderly was successfully demonstrated. Overwhelmingly positive user feedback, including direct inquiries about product availability, strongly validates its relevance and perceived value.

Currently, the SmartScan prototype is a hardcoded demonstration, validating the core concept with pre-recorded texts and basic interactions. For future development, the primary goal is a fully functional SmartScan. This ambitious step envisions integrating a real camera for live text capture, an AI-powered simplification engine, a specialized eye-friendly display, professionally engineered hinges, and continuous design iterations. These advancements are crucial for transforming SmartScan into a market-ready product that could significantly enhance the autonomy and quality of life for older individuals.

Beyond technical advancements, future SmartScan iterations aim to broaden the scope of its user base and integrate into wider assistive ecosystems. While the current prototype primarily addressed general literacy challenges, future adaptations could cater to more specific needs. For instance, simplified language output could greatly benefit non-native speakers or users with early-stage dementia by clarifying complex instructions and reducing cognitive load. Additionally, dynamic text-to-speech with customizable speed and accents could significantly increase accessibility for users with diverse visual impairments, allowing for a more personalized and effective auditory experience. In a broader, long-term perspective,

Figure 9: Prototype 1 after Iteration

Figure 10: High-Fidelity Prototype

SmartScan could potentially integrate into wider assistive ecosystems, such as care planning platforms or smart home (IoT) solutions. By enabling data exchange via standardized APIs, SmartScan might facilitate context-aware interventions like automated reminders or simplified medication instructions, creating a more connected support network.

To evaluate SmartScan's real-world impact, future studies would ideally measure usability using standardized methods like the System Usability Scale (SUS). Key metrics such as reading time reduction, comprehension accuracy, and user retention would be tracked to assess long-term effectiveness.

7 Conclusion

SmartScan emerged from a user-centered design process, proving how intelligent UX design can create transformative assistive AI solutions. Conceived as a simple, familiar-looking device, SmartScan directly addresses the challenge of reading small, complex texts by providing AI-powered simplification in the background. It effectively translates smartphone-like functionalities into a dedicated, user-friendly device designed to serve a single purpose without overwhelming the user.

The project's success is underscored by the enthusiastic reception from our target demographic. Not only did our commercial TV spot resonate as highly entertaining, but it also genuinely conveyed the product's tangible value to those at the "digital aid" center. This strong validation confirms that our approach successfully navigated initial skepticism towards AI by offering unobtrusive, beneficial technology. SmartScan demonstrates the feasibility of effective assistive AI for the elderly, highlighting the critical role of empathy and deep user understanding in designing solutions that truly lower barriers.

8 Work Distribution

The successful realization of the SmartScan prototype was a collaborative effort leveraging the specific strengths of each team member. Each team member was included in the interviewing, ideation, design and the low-fidelity prototyping process. Everyone had a hand in creating the storyboards and was included in design decisions. Louis and Amy were primarily responsible for the craftsmanship and aesthetic design of the high-fidelity prototype. This included conceptualizing its external appearance, integrating the display, and utilizing manufacturing techniques such as 3D printing and laser cutting, ensuring the product was visually appealing and functionally aligned with its analog design. Gaman and Lisa handled the technical programming of the prototype. Their responsibilities included choosing the right hardware, implementing the logic for displaying texts, controlling speaker output, and managing button press events as well as making a connection between the Raspberry and the display and the feedback of the LED string. Additionally, connecting everything together with wires and soldering. This division of labor enabled efficient development and integration of both the hardware and software aspects of the prototype.

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